Systematic Map Codebook

ILRI / Bristlepine Resilience Consultants — 2025–2026

Quick reference

Field	Valid values
`confirmed_include`	`yes` `no`
`publication_type`	`journal_article` `report` `working_paper` `thesis` `other`
`geographic_scale`	`local` `sub-national` `national` `multi-country` `regional`
`producer_type`	`crop` `livestock` `fisheries_aquaculture` `agroforestry` `mixed` `undefined`
`marginalized_subpopulations`	`women` `youth` `people_with_disabilities` `landless` `indigenous_peoples` `ethnic_minorities` `migrant_seasonal_workers` `other` `none_reported`
`adaptation_focus`	**free text**
`process_outcome_domains`	`knowledge_awareness_learning` `decision_making_planning` `uptake_adoption` `behavioral_change` `participation_coproduction` `institutional_governance` `access_information_services` `yields_productivity` `income_assets` `livelihoods` `wellbeing` `risk_reduction` `resilience_adaptive_capacity`
`indicators_measured`	**free text**
`methodological_approach`	`qualitative` `quantitative` `participatory` `modeling_with_empirical_validation` `experimental`
`purpose_of_assessment`	`research` `project_learning` `program_evaluation` `donor_reporting` `national_reporting`
`data_sources`	`surveys` `administrative_data` `remote_sensing` `participatory_methods` `secondary_data` `other`
`temporal_coverage`	`cross_sectional` `seasonal` `longitudinal` `repeated_cross_sectional`
`cost_data_reported`	`yes` `no`
`strengths_and_limitations`	**free text**
`lessons_learned`	**free text**

`confirmed_include`

Does this paper meet all five inclusion criteria?

Value	Meaning
`yes`	Paper meets all five criteria — proceed to fill in all 16 fields
`no`	Paper fails one or more criteria — leave remaining fields blank

Use the inclusion criteria below to make this decision. If you are unsure, enter your best judgement and add a note in the `notes` column.

Inclusion criteria

1. Population

Include if: Individuals or households engaged in small-scale agricultural production (crop, livestock, fisheries/aquaculture, agroforestry) in an LMIC. Also include marginalized producer groups (women, youth, landless farmers, indigenous peoples, ethnic minorities, migrant/seasonal workers).

Exclude if: Large-scale commercial agriculture, agribusiness without smallholder focus, or non-agricultural livelihoods only.

When in doubt:

Scenario	Decision
Mixed samples (smallholders + larger farms)	INCLUDE — if smallholders are mentioned and analyzed as a distinct group
Generic "Farmers" label in LMIC contexts	INCLUDE — unless study clearly focuses on commercial-scale operations
Industrial or corporate agriculture	EXCLUDE

2. Concept (Adaptation)

Include if: Study assesses or documents adaptation processes (learning, decision-making, adoption, participation, behavior change) OR adaptation outcomes (productivity, income, livelihoods, wellbeing, resilience, risk reduction).

Exclude if: - Climate impacts/vulnerability assessment ONLY (no adaptation action measured) - Mitigation only (reducing emissions) - General development or agronomy without explicit climate risk framing - Purely theoretical or conceptual (no intervention or measurement described)

When in doubt:

Scenario	Decision
Adoption/uptake of climate-smart agriculture or water management	INCLUDE
Livelihood diversification in response to climate stress	INCLUDE
Vulnerability index or risk score without coping/adaptation responses	EXCLUDE
Farmer awareness/perception of climate change alone	EXCLUDE

3. Context (Climate + Agriculture)

Include if: Agricultural setting (crop, livestock, fisheries, agroforestry) with explicit climate hazard link — drought, flooding, temperature extremes, rainfall variability, or other climate stressors affecting production.

Exclude if: - Non-agricultural sector - Agricultural study with no climate dimension - Climate mentioned only as background (not driving the research question) - Agricultural impacts/responses unrelated to climate

When in doubt:

Scenario	Decision
“Water scarcity” or “aridity” in semi-arid regions (implies climate)	INCLUDE
“Food insecurity” in LMIC rural contexts with adaptation responses	INCLUDE
Generic “livelihood change” in non-climate context	EXCLUDE
Agriculture + environmental change (without climate specificity)	EXCLUDE

4. Methodology

Include if: Study describes, evaluates, or applies a measurable method or framework to assess adaptation (indicators, metrics, surveys, qualitative interviews, participatory assessments, models with empirical data). Published 2005 or later.

Exclude if: - Purely theoretical/conceptual (no method or data) - Systematic reviews, meta-analyses, or literature reviews - Pre-2005 publication - Opinion pieces or policy briefs without empirical method

When in doubt:

Scenario	Decision
Narrative case study of farmer adaptation	INCLUDE — qualitative empirical method
Agent-based model with survey calibration	INCLUDE — model + empirical data
Pilot application of an adaptation tool	INCLUDE
Crop variety trial without farmer adoption analysis	EXCLUDE

5. Geography

Include if: Study conducted in one or more Low- or Middle-Income Countries (LMICs) or Global South countries. Single or multi-country studies OK if at least one LMIC is involved.

Exclude if: OECD high-income countries only (USA, UK, EU, Australia, Canada, Japan, South Korea, etc.) with no LMIC component.

When in doubt:

Scenario	Decision
China (all provinces)	INCLUDE — upper-middle-income
South Africa	INCLUDE — upper-middle-income
Multi-country study (e.g. Peru + Germany)	INCLUDE — Peru qualifies
Brazil, Vietnam, Thailand, Iran	INCLUDE — all are LMIC

The 16 coding fields

1. `publication_year`

Year the study was published. If multiple versions exist, use the most recent.

Valid values: integer — e.g. `2021`

Where to look: The publication year is typically listed at the very top or bottom of the pages, alongside the article's title, volume, and page numbers.

2. `publication_type`

Value	Meaning
`journal_article`	Peer-reviewed journal
`report`	Project/programme report, evaluation report, policy brief
`working_paper`	Pre-publication working paper or discussion paper
`thesis`	PhD or master's dissertation
`other`	Book chapter, conference paper, dataset paper

Where to look:

- **Headers, Footers, and Margins:** Check the very top or bottom of pages for journal name, volume, issue, and page numbers—these immediately identify peer-reviewed journal articles.
- **Cover Pages (Grey Literature):** Non-peer-reviewed reports often feature a standalone cover page with the logos and names of publishing organizations (NGOs, World Bank, FAO, research institutes, government ministries).
- **Disclaimers and Acknowledgements:** Look at footnotes on the first page or inside covers. Working papers and discussion papers often include explicit disclaimers stating the research is preliminary, circulating for discussion, or lacks formal peer review.
- **Degree Requirements (Theses/Dissertations):** The title page will typically include text stating "A thesis/dissertation submitted in partial fulfillment of the requirements for the degree of [Master's/PhD]..." followed by the university name.

3. `country_region`

Country or countries where the study was conducted. Use a region label only if no countries are named.

Format: free text; separate multiple entries with semicolons

Examples: `Tanzania` | `Ghana; Mali; Burkina Faso` | `East Africa` | `Global`

Where to look: Scan the **article's title, abstract, introduction, and Methods / Study Area description**—authors typically state the study location early and clearly.

4. `geographic_scale`

Value	Meaning
`local`	Village, community, single site
`sub-national`	District, province, region within a country
`national`	One country
`multi-country`	Two or more specific named countries
`regional`	Broad region without country-level focus

Where to look: Check the **Methods / Study Area / Site Description section**—this is the most definitive place. Authors must clearly state whether their study focused on a single village, a district within a country, the entire country, multiple named countries, or a broad geographic region.

Boundary rules for common edge cases:

- **Local vs. sub-national:** If a study takes place across multiple villages within a single district, code `sub-national` — not `local`. Local means a single site or community.
- **Multi-country vs. regional:** If the authors name the specific countries involved (e.g., Kenya, Uganda, Tanzania), code `multi-country`. Reserve `regional` strictly for papers that assess a broad area (e.g., "East Africa", "West Africa") without a country-level focus.

5. `producer_type`

Select all that apply; separate with semicolons.

Value	Meaning
`crop`	Crop farming systems
`livestock`	Livestock farming systems
`fisheries_aquaculture`	Fisheries or aquaculture systems
`agroforestry`	Agroforestry systems
`mixed`	Explicitly mixed crop-livestock
`undefined`	Generic "smallholder farmers" with no production system specified

Where to look: Look in the Abstract and Introduction: Authors usually state their target population and main agricultural focus right away. If not, check the Methods / Study Area / Site Description sections: This is the most reliable place to look, as authors must describe the economic and agricultural baselines of the communities they are surveying.

Do not default to `undefined` based on the abstract alone. If the abstract uses the generic term "smallholder farmers," check the Methods / Site Description section and any baseline summary statistics tables for mentions of crop or livestock assets before coding `undefined`.

6. `marginalized_subpopulations`

Code only what is **explicitly stated** in the paper — do not infer. **Only code a group if the intervention specifically targets them, or if the study explicitly disaggregates its outcome results by that group.** Do not code a group simply because they appear in a baseline demographic table. Select all that apply; separate with semicolons.

Value	Meaning
`women`	Women explicitly targeted or gender-disaggregated analysis
`youth`	Young farmers or age-disaggregated outcomes
`people_with_disabilities`	Persons with disabilities explicitly named
`landless`	Landless households or tenure-insecure groups
`indigenous_peoples`	Indigenous communities named
`ethnic_minorities`	Ethnic minority groups named
`migrant_seasonal_workers`	Migrant or seasonal agricultural workers
`other`	Any other marginalised group explicitly named — add a note in `notes`
`none_reported`	No marginalised groups mentioned

Where to look:

- **Abstract and Introduction:** Authors often state upfront if their study specifically targets or evaluates a vulnerable subgroup.
- **Methods / Sampling / Study Area sections:** This section details exactly who was surveyed and how they were chosen. Look for descriptions of "purposive sampling" criteria—e.g., "deliberately selecting participants based on gender (if known) and/or youth."
- **Results and Descriptive Statistics (Tables):** Check baseline household characteristics tables and demographic summaries to identify which groups were explicitly studied.
- **Discussion and Conclusion:** Authors often highlight how interventions impact marginalized groups differently—e.g., "inherent resource inequities between men and women constrain adaptation."

7. `adaptation_focus` *(free text)*

The specific climate adaptation action, intervention, or practice the study tracks. **Write 5–15 words maximum.** Write a single synthesised phrase — do not list every strategy mentioned. If a paper covers multiple strategies, describe the dominant one.

Examples: - `crop varieties and genetics` - `water management and irrigation` - `climate information and advisory services` - `livelihood diversification (non-farm)` - `agroforestry` - `livestock management` - `financial strategies` - `social and community networking`

Where to look:

- **Abstract and Introduction:** Authors usually highlight the primary interventions or adaptation strategies they are investigating right away.
- **Methods / Interventions / Study Design sections:** This is where authors detail the specific strategies tested, promoted, or surveyed in the study.
- **Results and Descriptive Statistics (Tables):** Look for tables detailing adoption rates, summary statistics, or regression variables—these list the exact practices tracked.
- **Discussion and Conclusion:** If the study is a modeling or vulnerability assessment that doesn't test a specific intervention, authors will often propose an adaptation focus in these sections.
- **Appendix / Survey Instrument:** If the researchers include their questionnaire, it will show the exact adaptation options they asked farmers about.

Example: In the Malawi study, the "Sampling methods and agroecological experiments" section provides a categorized list of the exact methods farmers selected: integration of trees, soil fertility methods, crop diversification, and livelihood diversification (e.g., small livestock).

8. `process_outcome_domains`

What the study measures. Select all that apply; separate with semicolons.

Rule: Code only domains that are **explicitly measured** — not domains that could be inferred. If the paper measures yield but not income, code `yields_productivity` only.

DO NOT use free text here. Enter only the valid code values from the table above. If a paper measures "food security," map it to `livelihoods`. If it measures "nutrition" or "dietary diversity," map it to `wellbeing`. Do not enter definitions, synonyms, or paraphrases.

Value	Meaning
`knowledge_awareness_learning`	Awareness of climate change, knowledge of options, learning outcomes
`decision_making_planning`	Farmer decision-making, use of forecasts in planning
`uptake_adoption`	Rates of adoption of a practice or technology
`behavioral_change`	Changes in farming or land management behaviour
`participation_coproduction`	Participation in programmes, co-design, farmer group engagement
`institutional_governance`	Policy uptake, institutional change, governance
`access_information_services`	Access to extension, credit, markets, climate information
`yields_productivity`	Crop yield, livestock productivity, fish catch
`income_assets`	Household income, asset accumulation
`livelihoods`	Food security, livelihood diversification
`wellbeing`	Health, nutrition, subjective wellbeing
`risk_reduction`	Reduced exposure or sensitivity to climate shocks
`resilience_adaptive_capacity`	Adaptive capacity indices, resilience scores, vulnerability indices

Where to look:

- **Methods / Variable Descriptions / Summary Statistics (Tables):** Authors will list the specific process indicators they tracked in these sections.
- **Results / Findings sections:** Qualitative and quantitative changes in knowledge, learning, and participation are reported here.
- **Discussion and Conclusion:** Authors often interpret how an intervention affected social processes, learning, or institutional dynamics in these sections.

Example: In the Central Vietnam study, the results section explicitly reports on farmers' "capacity to learn," "capacity to decide," and "capacity to act" in a dedicated results subsection.

9. `indicators_measured` *(free text)*

The specific indicators or metrics used. Extract from the methods or results sections. Include units where stated.

Examples: - `yield (t/ha), income (USD/season), food security score (HFIAS)` - `adoption rate (%), area under improved variety (ha)` - `climate resilience index score`

Where to look:

- **Tables (Variable Descriptions and Summary Statistics):** Authors almost always consolidate their indicators into comprehensive tables for clarity.
- **Methodology / Data Collection / Variables sections:** The narrative text in these sections describes the specific tools, scales, or census data used to capture the indicators.
- **Appendix / Survey Instrument:** If you need to see the exact questions asked to construct an indicator, the appendix is the most reliable place. For example, the Northern Ghana study includes "Appendix B: Survey questionnaire with indicators scoring criterion," which shows the precise multiple-choice questions used and how points were assigned to measure economic resources, social capital, and technology.

10. `methodological_approach`

Select all that apply; separate with semicolons.

Value	Meaning
`qualitative`	Qualitative research design (interviews, focus groups, ethnography)
`quantitative`	Quantitative research design (surveys, statistical analysis)
`participatory`	Only when participation is the **primary** design (PRA, photovoice) — not when farmers are merely surveyed
`modeling_with_empirical_validation`	Crop/climate/agent-based models validated with field data
`experimental`	RCTs, natural experiments, quasi-experimental approaches

Where to look:

- **Methods / Methodology / Materials and Methods section:** Authors typically dedicate this entire section to explaining their research design, often breaking it down into specific subsections.
- **Data Collection / Sampling subsections:** Here, authors detail whether they used surveys, expert interviews, focus groups, or secondary data.
- **Data Analysis / Empirical Approach subsection:** This subsection explains the statistical or qualitative frameworks used to evaluate the data.
- **Abstract:** Authors almost always provide a concise, one-to-two-sentence summary of their methods right at the beginning of the paper.
- **Introduction (final paragraphs):** Right after stating their research objectives at the end of the introduction, authors frequently include a brief preview of their methodology.

Example: The Malawi study introduces its methodology by stating: "A participatory research approach combined with pre-post longitudinal study design was used to examine changes..."

11. `purpose_of_assessment`

Value	Meaning
`research`	Academic study with no programme affiliation
`project_learning`	M&E embedded in an ongoing project
`program_evaluation`	Formal evaluation of a completed programme
`donor_reporting`	Report produced for a donor or funder
`national_reporting`	Contributing to NAPs, NDCs, or national monitoring frameworks

Where to look:

- **Abstract:** Authors typically provide a high-level summary of why the study was conducted right at the beginning of the document.
- **Introduction (end of section):** Standard academic writing conventions dictate that authors explicitly outline their goals, objectives, or research questions at the very end of the introduction section, right before diving into the methodology. **Example:** "The specific research objective of this study is to test whether agroecological farming methods... can improve food security..."
- **Conclusion / Summary:** Authors often briefly restate the core purpose of their assessment in the concluding section when summarising what their study ultimately set out to achieve and what it found.

12. `data_sources`

Select all that apply; separate with semicolons.

DO NOT use free text here. Enter only the valid code values from the table below. Note the underscores — `secondary_data` not "secondary data". For mixed-methods studies, code **all** sources used, not just the first one you find.

Primary vs. secondary data: Primary data is original information collected directly for the study (household surveys, expert interviews, participatory focus groups). Secondary data is pre-existing information the authors did not collect themselves (existing datasets, literature reviews, government records). Interviews and focus groups are primary data — map them to `surveys` or `participatory_methods` as appropriate.

Value	Meaning
`surveys`	Primary household or farm surveys, structured interviews
`administrative_data`	Government or programme records
`remote_sensing`	Satellite or aerial imagery
`participatory_methods`	PRA, focus groups, photovoice as a data source
`secondary_data`	Existing datasets, literature, or other secondary sources
`other`	Any data source not listed above

Where to look:

- **Methods / Methodology / Materials and Methods section:** This is the primary and most detailed location. Authors almost always include specific subsections dedicated to explaining where their data came from. **Example:** In the Northern Ghana study, the "Data collection" subsection under "Research methods" explicitly outlines the use of primary data (expert interviews and structured questionnaires) and secondary data (a systematic literature review).

- **Data / Data and Variables sections:** Sometimes, authors dedicate a top-level section specifically to the data used, especially in economic or quantitative studies relying heavily on existing secondary datasets or household panels. **Example:** The East Africa study features a section titled "Data, description of variables, and summary statistics," which explains that their data came from a baseline household survey collected by the CGIAR Research Program on Climate Change, Agriculture, and Food Security (CCAFS).

13. `temporal_coverage`

Value	Meaning
`cross_sectional`	Single point in time or single season
`seasonal`	Intra-annual, across one or a few seasons
`longitudinal`	Multi-year, panel data, repeated measures with the same individuals
`repeated_cross_sectional`	Multi-year, repeated measures with different individuals each time

Where to look:

- **Abstract:** Authors frequently summarize the study's timeframe or experimental design right at the beginning. **Example:** The Malawi study abstract explicitly mentions a "four-year study," baseline and follow-up surveys conducted in 2011 and 2013, and the use of "Longitudinal mixed effects models."
- **Methodology / Data Collection / Study Design sections:** This is the most definitive place to find temporal coverage—authors must detail exactly when and how often they collected their data.
- **Data and Variables / Climate Modeling sections:** If the study relies on secondary data, historical records, or predictive modeling (rather than field surveys), these sections will define the exact time periods analyzed.

14. `cost_data_reported`

Value	Meaning
`yes`	Paper explicitly reports cost figures or cost-effectiveness ratios
`no`	No cost data — a qualitative mention of "limited resources" counts as `no`

Where to look:

- **Results / Economic Evaluation sections:** If the core purpose of the study involves an economic or financial assessment, the exact cost figures, cost-benefit analyses, or cost-effectiveness ratios will be reported in the results, usually accompanied by data tables.
- **Discussion and Conclusion:** Authors may summarize the exact financial requirements for implementing an adaptation strategy at a regional scale, or they may compare its cost-effectiveness against other measures.

Quick-search tip: Use your document viewer's search function to scan for currency symbols or acronyms (e.g., $, USD, VND, Euro) and economic keywords like "cost-effectiveness," "CBA" (cost-benefit analysis), "expenditure," or "budget."

15. `strengths_and_limitations` *(free text)*

Author-reported strengths and limitations. Look in the Discussion, Conclusion, and Limitations sections. Extract verbatim where possible.

Format: `Strength: <text>. Limitation: <text>.`

You must use this exact prefix format. Do not use parentheses, dashes, or other formats. Do not write two-word summaries (e.g., "Strength: Mixed methods." is not acceptable — quote or closely paraphrase the author's actual words).

Where to look:

- **Discussion Section:** This is the most common place to find both strengths and limitations. As authors interpret their findings, they typically reflect on the robustness of their study design, highlight how their research makes strong contributions to existing literature, and acknowledge any methodological, sample size, or data constraints.
- **Conclusion:** Authors often briefly summarize the main strengths (e.g., "This study provides novel insights into…") and note any boundaries to their findings, frequently tying these limitations directly to recommendations for future research.
- **Dedicated "Limitations" subsection:** Some journals require authors to include a specific subsection—usually at the very end of the Discussion—explicitly detailing the study's constraints.

16. `lessons_learned` *(free text)*

Key lessons or recommendations the authors report. 1–3 sentences. Focus on what they say should be done differently — not the findings themselves. Enter `not_reported` if absent.

Where to look: Look at the Discussion section: This is the primary area where authors interpret their results, explain why certain interventions worked or failed, and extract practical insights. The Conclusion (or specific "Implications" and "Recommendations" subsections): Authors consistently use the final sections of the paper to synthesise their overarching lessons for policymakers, donors, or future researchers. The Abstract (Final Sentences): Authors frequently distill their most critical lesson learned into the final lines of the abstract.

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